

Module 2: Overfishing

```
library(tidyverse)
```

Use of GitHub

Link to your forked GH repository: <https://github.com/nikhilshah7/CLES131-module1-climate>

Use of Quarto

Link to your .qmd file: <https://github.com/nikhilshah7/CLES131-module1-climate/blob/main/module1.qmd>

Overexploitation of global fisheries

We will examine the state of global fisheries and seek to reproduce one of the most widely cited examples of species collapse in order to examine the evidence behind an influential paper on global fisheries, [Worm et al. 2006](#). However, rather than using the 1950-2003 dataset available to Worm and colleagues in 2006, we will be drawing from more recent stock assessment data to see how the trends have fared, specifically in Atlantic cod.

We also explore how to understand and manipulate tabular data using relational database concepts. Instead of working with a single independent dataframe, as in Module 1, we will be working with a large relational database consisting of tables of different dimensions, but are related through one or more IDs.

The Database

We will use data from the [RAM Legacy Stock Assessment Database](#). Note that the database files are available through Zenodo; please download and unzip the file, moving the .RData file only into your own `data/` folder within this project.

Use the code below to load up the tables in this database.

```
load("data/DBdata[asmt][v4.66].RData")
```

The above function loads all 60 tables of this database, which is a lot! So step one of the data science process is exploring the structure of the database and figuring out how the tables are related.

A selected set of documentation have been provided in `documentation/`. Open up the Database Quickstart Guide for reference as you work through the problems below.

Q1 (1 point)

One of the main metadata tables is 'stock':

- How many stocks are there and how many species do they represent?

```
nrow(stock)
```

```
[1] 1514
```

```
# 1514
```

```
length(unique(stock$tsn))
```

```
[1] 398
```

```
# 398
```

- What are the top three most common stocks? (Okay to use common names; these are standardized for fish)

```
stock |>
  group_by(commonname) |>
  summarize(n_stocks = n()) |>
  arrange(desc(n_stocks))
```

```
# A tibble: 410 x 2
  commonname      n_stocks
  <chr>          <int>
1 Pink salmon      170
2 Sockeye salmon   104
3 Chum salmon       99
4 Norway lobster   38
5 Atlantic cod     28
6 Herring          24
7 Anchovy          19
8 Snow crab        19
9 Northern shrimp  17
10 Haddock          14
# i 400 more rows
```

```
# Pink salmon (170)
# Sockeye salmon (104)
# Chum salmon (99)
```

- In your own words, describe the unit of observation in the ‘stock’ table and some associated variables that describe it.
- Each row describes one fish stock, a population of a species of fish in one given location. Variables relate to the species of fish (tsn, scientificname, commonname), stock location (areaid, region, primary_country), and stock status (state).

Q2 (1 point)

The next main metadata table is ‘assessment’:

- What is the unit of observation in the ‘assessment’ table? How is this related to the unit of observation in the ‘stock’ table?

```
# Each row is one assessment of one stock.
```

```
length(unique(assessment$stockid))
```

```
[1] 1514
```

```
# Each stock is assessed at least once in this table (1514 stocks total). Stocks may be assessed
```

- It seems like ‘stockid’ and ‘assessyear’ might uniquely identify each assessment. Is this the case? If not, why not?

```
assessment |>
  group_by(stockid, assessyear) |>
  summarize(n = n()) |>
  arrange(desc(n))
```

‘summarise()’ has grouped output by ‘stockid’. You can override using the ‘.groups’ argument.

```
# A tibble: 3,449 x 3
# Groups:   stockid [1,514]
  stockid      assessyear      n
  <chr>        <chr>    <int>
1 ANCHIXa      1989-2016        2
2 BLINGVa-XIV  2000-2016        2
3 BRILL2232    1970-2016        2
4 COD1IN       1911-2016        2
5 CODBA2224    1993-2016        2
6 CODBA2532    1965-2016        2
7 CODIIaW-IV-VIIId 1962-2016        2
8 DAB2232      1970-2016        2
9 EFLOUN2223    1973-2016        2
10 EFLOUN2425    1973-2016        2
# i 3,439 more rows
```

No, some stocks were assessed multiple times for the same time period, such as anchovy in 1989-2016.

- Which variable(s) might you use to join ‘stock’ and ‘assessment’?
- stockid, as it seems to be the only shared variable between the two datasets.

Q3 (1 point)

In section B of the Database Quick Guide, additional metadata tables are listed in section B. With your data exploration skills, complete the following table:

B table	Associated table	join column from B	join column from other
area	stock	areaid	areaid

B table	Associated table	join column from B	join column from other
assessmethod	assessment	methodshort	assessmethod
assessor	assessment	assessorid	assessorid
management	stock	country	primary_country
taxonomy	stock	tsn	tsn

*Note that assessmethod is not a data.frame, which was perhaps an oversight; consider turning it into one for the ease of using tidyverse functions.

```
assessmethod <- data.frame(assessmethod)
```

Q4 (1 point)

There are two main data tables described in section C, but their key attributes are difficult to understand. Join the two main data tables with their respective metrics tables (section D):

```
bioparams_metrics <- bioparams |>
  left_join(biometrics, join_by(bioid == biunique))

timeseries_metrics <- timeseries |>
  left_join(tsmetrics, join_by(tsid == tsunique))
```

- Briefly describe the contents of each joined table.
- Biometrics adds detailed biological data according to the code that splits each stock into several sets of “demographics” related to ages, deaths, spawning, etc. Tsmetrics adds information about what is being measured to give the provided value for each assessment in each year.
- What is accomplished by joining these tables?
- The bioid/tsid codes aren’t really useful for obtaining usable information from the tables, so all the detailed information that that column encodes can be expanded and added to the tables if necessary by joining, which allows all that data to be found in one table while not using as much space.

Overexploitation of Atlantic cod

This database is clearly complex and contains a lot of information. We’ve grappled with its high dimensionality by exploring and joining tables above. Next, we will take a “bottom-up” approach and evaluate a single species, Atlantic cod. Read this [book chapter](#) on Atlantic cod, which serves as a reference for the following questions.

Q5 (1 point)

The collapse of the Atlantic cod fisheries is well documented. Let's see if we can visualize that decline with this database.

- How many unique fish stocks comprise “Atlantic cod” and what regions do they come from?

```
stock |>
  filter(commonname == "Atlantic cod") |>
  nrow()
```

```
[1] 28
```

```
# 28

stock |>
  filter(commonname == "Atlantic cod") |>
  group_by(region) |>
  summarize(count = n())
```

```
# A tibble: 4 x 2
  region          count
  <chr>          <int>
1 Canada East Coast    12
2 Europe non EU         6
3 European Union       8
4 US East Coast         2
```

```
# Canada East Coast (12)
# Europe non EU (6)
# European Union (8)
# US East Coast (2)
```

Create a vector of the ‘stockid’ associated with “Atlantic cod”. Then turn your attention to the previously-joined timeseries/tsmetrics table and create a subsetting dataframe with only observations of “Atlantic cod” that have a non-NA value. This table reports many years and many metrics, while we are primarily interested in the years since 1950.

```
codstock <- stock |>
  filter(commonname == "Atlantic cod")

codstockid <- codstock$stockid

cod_timeseries <- timeseries_metrics |>
  semi_join(codstock, by = 'stockid')
```

- Which metrics are most commonly reported for Atlantic cod in this timeframe and what do they mean?

```
cod_timeseries |>
  group_by(tslong) |>
  summarize(count = n()) |>
  arrange(desc(count))
```

```
# A tibble: 44 x 2
  tslong                                count
  <chr>                                <int>
1 Catch divided by mean catch          7442
2 General total Catch (TC then TL, MT units only) 7442
3 Total landings                       6278
4 Spawning Stock Biomass               5985
5 Recruits                            5692
6 Fishing mortality (preferred metric). 5367
7 Total catch (i.e. landings + discards. Add landings + discards to get ~ 4791
8 General fishing mortality or exploitation rate time series preferentia~ 4435
9 General fishing mortality or exploitation rate time series preferentia~ 4435
10 General exploitation rate time series 4368
# i 34 more rows
```

```
# Catch divided by mean catch (7442): Ratio between catch in this assessment and the mean
# General total catch (7442): Total amount of fish extracted from water in metric tons
# Total landings (6278): Total amount of fish brought to land and not discarded, portion of
# Spawning stock biomass (5985): Biomass of all fish in the area that are able to reproduce,
```

Q6 (1 point)

Make a timeseries figure for each stock of Atlantic cod. Rather than coloring by the identity of the stock, color by the region it comes from (this may require joining). Use `facet_*`() to

plot both metrics in a single code chunk. Connect the points for each stock and focus on the period since 1950. Update until the figure is easily interpretable.

```
regions <- stock |>
  select(stockid, region)

cod_timeseries_region <- cod_timeseries |>
  left_join(regions)
```

Joining with `by = join\_by(stockid)`

```
# only metrics: general total catch + total landings

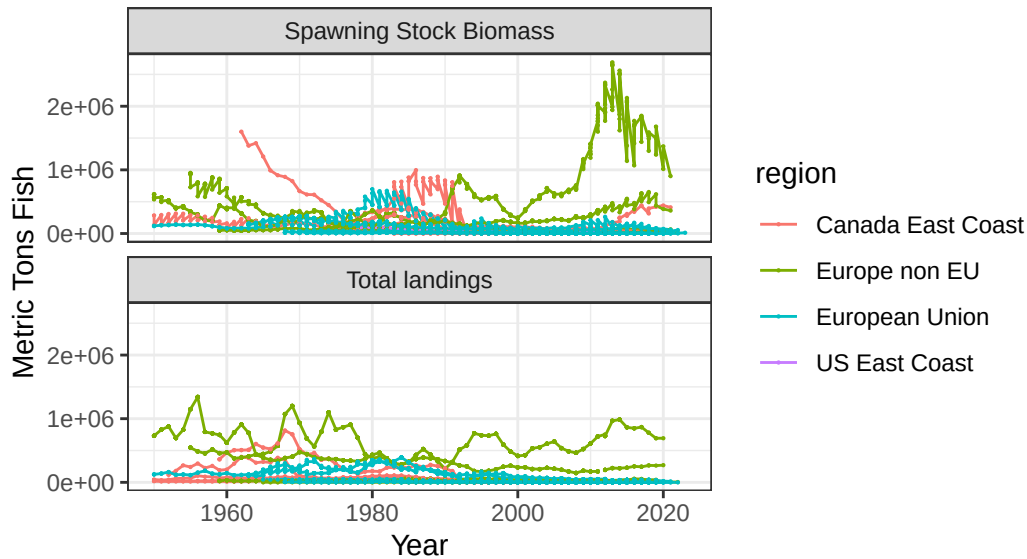
cod_timeseries_region |>
  filter(tsshort == 'TL' | tsshort == 'SSB') |>
  filter(tsyear >= 1950) |>
  ggplot(aes(x = tsyear,
             y = tsvalue,
             color = region)) +
  geom_line(aes(group = stockid)) +
  geom_point(size = 0.1) +
  facet_wrap(~tslong,
            nrow = 2) +
  labs(x = 'Year',
       y = 'Metric Tons Fish',
       title = 'Timeseries of fish biomass in Atlantic Cod stocks',
       subtitle = 'Total biomass of spawning fish and biomass of fish brought to land') +
  theme_bw()
```

Warning: Removed 98 rows containing missing values or values outside the scale range (`geom\_line()`).

Warning: Removed 120 rows containing missing values or values outside the scale range (`geom\_point()`).

Timeseries of fish biomass in Atlantic Cod stocks

Total biomass of spawning fish and biomass of fish brought to land



- What happened to Atlantic cod stocks during the period 1950-present? What information does each metric convey?
- Atlantic cod stocks have varied significantly over the measured timeframe, but aside from those in the 'Europe non EU' region, which mostly concerns near-polar latitudes, a notable decline can be observed in the late 20th century, with North American stocks declining to almost zero biomass and landings around 1990. Fishing of Atlantic cod in North America seems to have stopped since then, and a resurgence in cod population occurred in the 2010s.
- Spawning stock biomass shows the amount of fish present in the stock (which are capable of reproduction, which can help to indicate if the stock will be sustained in the future). Total landings shows the amount of fish that are removed from the water and not returned, which can help to show the effects of overfishing and the consequences of low fish stock on fish availability.

```
# shows places in 'Europe non EU' where atlantic cod stocks are assessed
stock |>
  filter(region == 'Europe non EU' & commonname == "Atlantic cod") |>
  distinct(stocklong)
```

```
stocklong
1 Atlantic cod NAFO 1F and ICES 14
2 Atlantic cod Faroe Bank
3 Atlantic cod Faroe Plateau
```

```
4 Atlantic cod Iceland Grounds
5 Atlantic cod North-East Arctic
6 Atlantic cod North-East Arctic (Norwegian coastal waters)
```

Q7 (1 point)

We will recreate a slightly more complex version of a figure from the [Millennium Ecosystem Assessment](#) (Fig. 11).

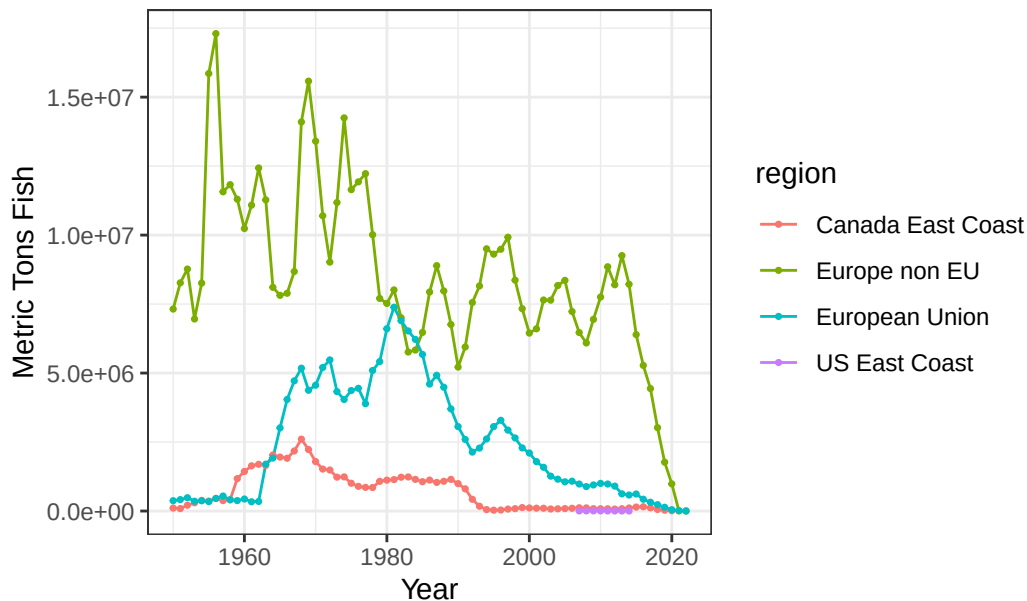
Selecting a metric that can be added across stocks, calculate the total fish landings for each region of Atlantic cod. Make a plot similar to Fig. 11 and color by region.

```
cod_timeseries_regional <- cod_timeseries_region |>
  filter(tsshort == 'TL') |>
  filter(tsyear >= 1950) |>
  group_by(region, tsyear) |>
  summarize(region_metric = sum(tsvalue, na.rm = T)) |>
  mutate(metric = 'Total Landings')
```

`summarise()` has grouped output by 'region'. You can override using the `.groups` argument.

```
cod_timeseries_regional |>
  ggplot(aes(x = tsyear,
             y = region_metric,
             color = region)) +
  geom_line() +
  geom_point(size = 0.6) +
  labs(x = 'Year',
       y = 'Metric Tons Fish',
       title = 'Timeseries of fish landings in Atlantic Cod stocks') +
  theme_bw()
```

Timeseries of fish landings in Atlantic Cod stocks



Which region does Fig. 11 from the MEA correspond to? - Newfoundland → Canada East Coast

Q8 (1 point)

Under the Management Policies section of the [book chapter](#), the authors summarize a few major efforts to restore and protect stocks of Atlantic cod. Focusing only on the US and Canada, update your plot from above to include vertical lines to represent the timing of least three major events and label them clearly on your plot. Doing so will likely involve creating a separate dataframe and using it in one of your ggplot layers (one possible argument within `aes()` is `linetype`). Facet the plot by region and update until it is easily interpretable.

- In 1976, US Congress passed a law that would restrict foreign fishing. Federal and regional agencies imposed regulations and fish management strategies to protect fish stocks. Amendments to strengthen this law were passed in 1996 and 2016. Cod was designated as a vulnerable species in 1996. Regulations focus on limiting access to critical waters and setting caps on the amount of fish that can be caught.

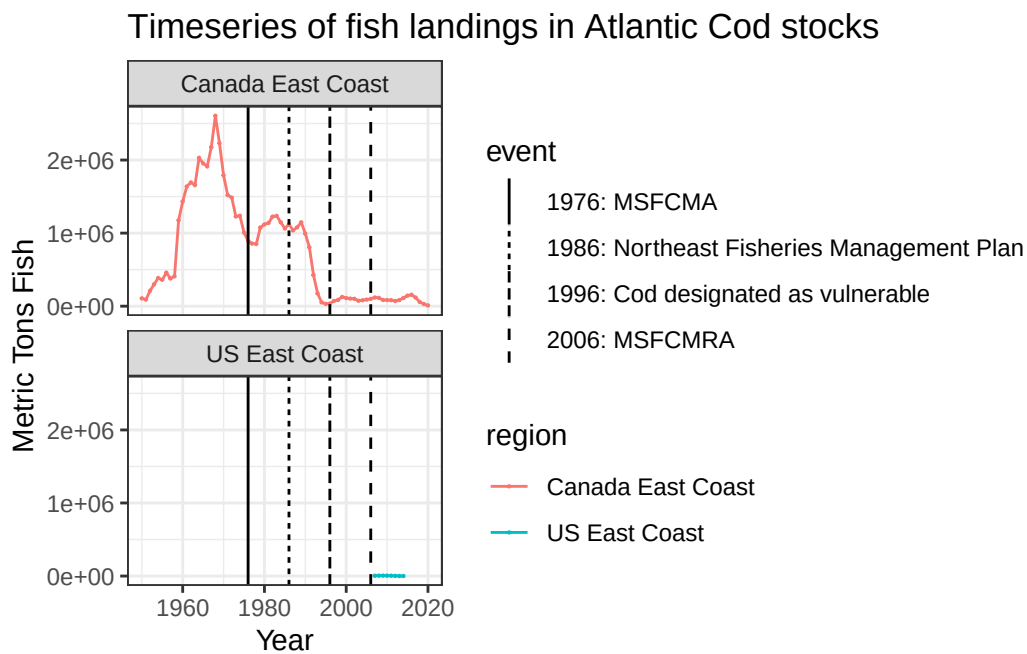
```
events <- data.frame(event = c('1976: MSFCMA', '1986: Northeast Fisheries Management Plan',
                              year = c(1976, 1986, 1996, 2006))

cod_timeseries_regional |>
  filter(region == 'Canada East Coast' | region == 'US East Coast') |>
```

```

ggplot(aes(x = tsyear,
           y = region_metric,
           color = region)) +
  geom_line() +
  geom_point(size = 0.1) +
  geom_vline(data = events,
            aes(xintercept = year,
                linetype = event)) +
  facet_wrap(~region,
            nrow = 4) +
  labs(x = 'Year',
       y = 'Metric Tons Fish',
       title = 'Timeseries of fish landings in Atlantic Cod stocks') +
  theme_bw()

```



Q9 (2 points)

What happened to Atlantic cod stocks for each region following each piece of legislation or management plan, and why do you think this was so? Your multi-paragraph answer should be informed by content from the book chapter; additional sources are optional, and all sources should be cited.

```

events <- events |>
  rbind(data.frame(event = '1992: Canadian moratorium',
                    year = 1992))

cod_timeseries_regional_spawn <- cod_timeseries_region |>
  filter(tsshort == 'SSB') |>
  filter(tsyear >= 1950) |>
  group_by(region, tsyear) |>
  summarize(region_metric = sum(tsvalue, na.rm = T)) |>
  mutate(metric = 'Spawning Stock Biomass') |>
  bind_rows(cod_timeseries_regional)

```

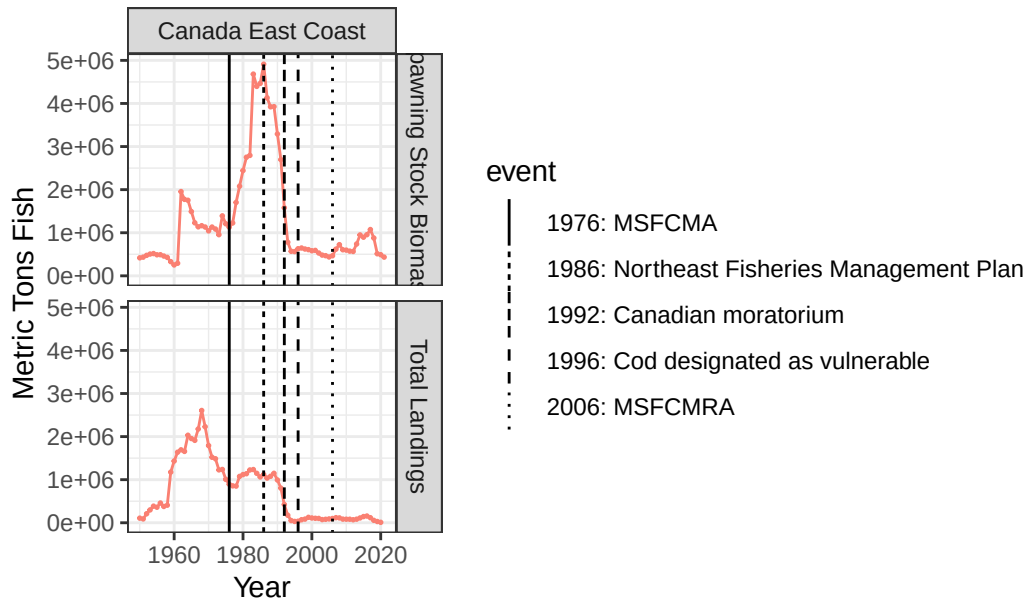
`summarise()` has grouped output by 'region'. You can override using the
 `.groups` argument.

```

cod_timeseries_regional_spawn |>
  filter(region == 'Canada East Coast') |>
  ggplot(aes(x = tsyear,
             y = region_metric)) +
  geom_line(color = 'salmon') +
  geom_point(size = 0.3,
            color = 'salmon') +
  geom_vline(data = events,
            aes(xintercept = year,
               linetype = event)) +
  facet_grid(rows = vars(metric),
            cols = vars(region)) +
  labs(x = 'Year',
       y = 'Metric Tons Fish',
       title = 'Timeseries of spawning stock biomass in Atlantic Cod stocks') +
  theme_bw()

```

Timeseries of spawning stock biomass in Atlantic Cod stocks



While the dataset seems to be lacking a lot of data for the US East Coast, it is essentially complete for Canada East Coast, which has experienced a tumultuous series of events related to its fish stock in the last fifty years.

In the mid-1970s, roughly when US Congress passed the Magnuson-Stevens Fishery Conservation and Management Act (Brownell et al., 2023), it became evident that large volumes of fishing of Atlantic cod could pose a problem for local fish stocks, so the Canadian government instated a similar ban on foreign fishing within Canadian waters (MacDowell, 2012), which promoted a significant rise in fish stocks. However, this simply allowed local fishing to supplant foreign fishing, and landing numbers stayed the same, even under scientific advisories to reduce the number of fish allowed to be caught. Around 1990, stocks collapsed heavily. Canada placed a moratorium on cod fishing in 1992 (MEA, 2005), which led to a drop in landings to near-zero, but this did not induce a recovery in fish stocks for several decades.

The authors of *Extinction Stories* suggest that overfishing may have led to consolidation of the resulting groups of fish (Brownell et al., 2023), and the decreased amount of groups made stocks extremely vulnerable to further fishing targeting the new, large groups, even when staying within fishing limits. This extreme reduction in fish population would have made it very difficult for fish populations to recover as fast as fishing regulation agencies might have expected or hoped for. Even 30+ years after the moratorium was instated, cod populations have failed to recover consistently to pre-1970s levels. This exemplifies how extreme, poorly-managed overfishing for even a short period of time can collapse fish stocks for several decades.

MacDowell, L. (2012). *An Environmental History of Canada*. Vancouver UBC Press.